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SE 216- SOFTWARE PROJECT MANAGEMENT
Spring 2025-2026

Monitoring and Assistance System for At-Home Physiotherapy Exercises

Problem Definition

Today, physiotherapy plays a vital role in the recovery process of many patients of all ages recover from injury, surgery or mobility limitations. After examination, doctors and physiotherapists usually give a series of exercises that patients should perform regularly at home. However, many patients find it very difficult to do these exercises correctly without a professional watching them. Some patients may skip exercises, while others may perform the movements incorrectly. Incorrect movements can reduce the effectiveness of treatment and, in some cases, even lead to new injuries.

Patients are unlikely to visit a physiotherapy facility on a daily basis because it can be expensive and time-consuming. For instance, it might not be possible for elder people to go to a physiotherapy center every day. In a similar way, it is not feasible for physiotherapists to use video calls to continuously monitor every patient. As a result, there is a gap between the workouts that patients actually undertake at home and those that doctors recommend.

Our project aims to create a technology-based mobile application that will help patients perform physiotherapy exercises more precisely and safely in a home environment. The mobile application will be used by the patients through a gadget such as a tablet or a mobile phone, where they will be able to see the exercises performed through the presentation of the exercises through pre-recorded videos and customized 3D figures.

The system will be able to analyze the movements of the patients through the camera of the gadget. It will identify the movements performed incorrectly by the patients. The physiological data of the patients, such as heart rate, will be collected using a sensor band. The system will be able to send a progress report to the physiotherapist. This will be possible through the analysis of the accuracy of the movements performed by the patients. The movements will be analyzed to identify the difficulties faced by the patients. This will be a great contribution to the rehabilitation of the patients.

Background Information

Physiotherapy exercises are important for patients' recovery. However, many patients may perform the exercises incorrectly or not practise regularly without professional supervision at home. Access is also limited by distance, time and cost issues. Technology-based solutions can make exercises safer and more effective while enabling physiotherapists to monitor progress remotely.

Objectives

- Help patients to complete their physiotherapy exercises at home:

  Reducing the need for doctor or clinic appointments and saving time.

- Improve access to physiotherapy services:

  Provide support to those living in rural areas or far from hospitals, where regular access to physiotherapy may be limited.

- Reduce the risk of incorrect movements:

  Identify and alert users to incorrect movements during exercises to avoid potential injuries.

- Encourage patients to exercise often:

  Encourage patients to follow their physiotherapy programmes regularly.

- Track exercise performance and consistency:

  Throughout the rehabilitation process, monitor how often and how accurately patients do their exercises.

Approval Signatures and GitHub Accounts

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